

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 04: Enterprise Architecture for E-Commerce

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Session plan (90 minutes)

- ① Why enterprise architecture matters in e-commerce
- ② Enterprise architecture as a set of views
- ③ Capability maps
- ④ Value streams
- ⑤ Domain boundaries and bounded contexts
- ⑥ Managerial implications

Learning objectives

By the end of this lecture, you should be able to:

- Explain how enterprise architecture (EA) helps align business goals, operating models, and technology decisions in e-commerce.
- Use capability maps, value streams, domain boundaries, and system-of-record thinking to structure a commerce platform.
- Distinguish current-state architecture from target architecture and transition planning.
- Analyze how AI capabilities fit into enterprise architecture without turning the architecture into a collection of isolated point solutions.

Why enterprise architecture matters in e-commerce

- business strategy,
- organizational capabilities,
- application boundaries,
- data ownership,
- integration patterns,
- and technology constraints.

Enterprise architecture as a set of views

- **Business architecture:** capabilities, value streams, organization roles, policies, and business outcomes.
- **Application architecture:** major systems, services, platform modules, ownership boundaries, and interactions.
- **Data architecture:** entities, identifiers, systems of record, analytical pipelines, and governance rules.
- **Technology architecture:** hosting environment, integration infrastructure, observability stack, identity/security, and platform services.

- **Core e-commerce capabilities:** customer acquisition and campaign orchestration,; product information and catalog management,
- **Why capability maps are useful:** Is personalization a standalone capability, or is it part of a broader discovery capability?; Which parts of the recommendation workflow belong to the storefront experience layer, and which belong to a shared data platform?

- **Typical e-commerce value streams:** **Browse-to-buy:** discovery, consideration, cart, checkout, payment, confirmation.; **Order-to-cash:** order capture, validation, fulfillment, invoicing, settlement, financial posting.
- **Why value streams matter architecturally:** Value streams expose cross-domain handoffs.

Domain boundaries and bounded contexts

- **Typical commerce domains: Catalog:** products, attributes, categories, media, content enrichment.; **Pricing and promotions:** price lists, discount rules, coupon logic, campaign offers.
- **Good boundaries versus bad boundaries:** pricing rules duplicated in storefront, ERP, and marketing tools,; customer attributes updated independently in CRM and commerce applications,

Systems of record and systems of engagement

- **Typical patterns:** The **commerce layer** often owns high-velocity interaction data such as carts, storefront sessions, and recommendation exposures.; The **ERP layer** often owns invoices, financial postings, stock movements, supplier records, and contractual B2B customer data.
- **Systems of engagement:** Storefronts, mobile apps, support consoles, and seller portals are often systems of engagement rather than systems of record.

Reference architecture for AI-driven e-commerce

- **Where AI fits:** search ranking and recommendations within discovery,; fraud scoring within checkout and payments,

Current state, target state, and transition architecture

- **Current-state analysis:** duplicated systems and overlapping responsibilities,; brittle integrations and manual reconciliation steps,
- **Target architecture:** Which domains and systems will own which capabilities?; Which integrations will be synchronous, asynchronous, or batch?
- **Transition architecture:** separating catalog and pricing ownership before replacing checkout,; introducing a shared event stream before centralizing experimentation,

Architectural concerns specific to AI

- **Training-serving consistency:** features must mean the same thing offline and online.
- **Latency budgets:** recommendation or fraud decisions may need sub-second responses.
- **Fallback behavior:** the commerce workflow must continue safely if a model is unavailable.
- **Observability:** systems must track not only uptime but also prediction quality, drift, and business impact.
- **Governance:** model changes may alter business policy and therefore require approval, auditability, and rollback procedures.

Operating model and team topology

- **Typical team patterns:** **Domain-aligned product teams:** own specific business capabilities such as checkout or catalog.; **Platform teams:** provide shared infrastructure such as identity, observability, experimentation, or data platforms.

Artifacts for students and architects

- a capability map,
- one or more value stream maps,
- a domain map with ownership boundaries,
- a system-context diagram,
- a system-of-record matrix for key entities,
- and a target-state architecture note with transition priorities.

Managerial implications

- slow product delivery because every change triggers cross-team negotiation,
- financial reconciliation issues from conflicting commercial truth,
- duplicated AI initiatives that solve the same problem inconsistently,
- and weak governance when no domain clearly owns a high-impact decision.